Hankel-Structured Channel Estimation for Magnitude-Only Rydberg Atomic Receivers: When Does Structure Help?

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***Abstract*—The magnitude-only Rydberg receiver model studied here measures the strength of a received field without directly observing its phase. This makes complex channel estimation difficult. We investigate when the spatial structure of a multipath channel helps this estimation problem. We propose a Hankel-structured extension of expectation–maximization Gerchberg–Saxton, called HS-GS, which uses the relationship between channel coefficients across a uniform linear array. The rank is chosen from held-out pilot measurements, and all pilots are then used for the final estimate. Selecting the maximum rank returns the estimator to EM-GS. Simulations show that HS-GS is most useful when the channel is simple relative to the array's structural capacity. However, smooth singular-value filtering can outperform hard-rank processing, including cases where HS-GS gives little benefit. The useful region therefore depends on how the structure is applied. We also examine reference strength, the pilot split and array position errors. Effective rank helps explain the results, while a version estimated from measurements remains sensitive to noise. These findings describe the benefits and limits of the tested structural methods without assuming a universal complexity threshold.**

***Index Terms*—Rydberg atomic receiver, channel estimation, phase retrieval, Hankel matrix, constrained Cramér–Rao bound, effective rank.**

I. Introduction

SIXTH-GENERATION wireless systems are expected to operate over wider bandwidths and at increasingly high carrier frequencies, placing new demands on receiver sensitivity and reconfigurability. Quantum sensing has emerged as one possible response, and Rydberg atomic receivers have recently attracted attention for wireless communications [1], [2]. These receivers use laser-excited atoms to sense an incident radio-frequency field through changes in optical transmission. In the magnitude-only readout considered here, the receiver observes the field amplitude rather than its full complex value, so the channel phase is not directly observed.

Recovering a complex channel from these amplitude-only measurements is a phase-retrieval problem. A known reference field provides phase-sensitive information by converting relative phase into observable amplitude. Recovery still depends on the reference and pilot design. Existing approaches include iterative phase-retrieval methods [3], gradient descent with structural projection [4], and unrolled networks [5]. These works mainly focus on improving channel estimation performance. Less attention has been given to a more basic question: when does imposing channel structure actually help, and when does that structure become too restrictive? This paper addresses that question using the spatial structure of multipath channels on a uniform linear array.

Related Rydberg-array work also uses recovered channel information for sensing. Quantum-MUSIC [6] reconstructs the channel matrix from magnitude measurements before applying MUSIC for multiuser sensing. Our endpoint is channel NMSE and the cost of spatial structural processing, rather than direction-of-arrival accuracy. We therefore use this work to place the problem in context, not as an interchangeable channel-estimation benchmark.

Low-rank Hankel modeling and Cadzow processing are established tools [7], [8]. Related uses of structured low-rank matrices include spectral compressed sensing and MRI reconstruction [9], [10], [11]. Our contribution is not a new general low-rank recovery principle. We study how a simple spatial Hankel correction behaves inside a magnitude-only Rydberg channel estimator, including conditions where it gives little benefit or causes harm. The Hankel matrix is formed along each user's array dimension. This construction differs from a rank constraint on a two-dimensional channel slice, as considered in [4].

Hard truncation gives a best rank-limited approximation to its input matrix in the Frobenius norm. It need not give the best estimate of the underlying clean channel. Singular-value shrinkage methods such as OptShrink address this distinction [12]. Their matrix-noise assumptions require care here: Hankel lifting repeats entries, and the error entering the structural step comes from an iterative channel estimate. We therefore do not assume that a shrinkage rule is optimal for this problem simply because it is optimal under a different denoising model.

We propose a Hankel-structured extension of EM-GS for magnitude-only channel estimation. HS-GS uses the spatial relationship between the channel coefficients, selects a rank from held-out pilots, and returns to EM-GS when the maximum rank is selected. We study how its benefit changes with array size, path count and effective rank. We also compare it with other structural processing, examine reference strength and pilot splitting, and test mild array-position errors. These comparisons show that the useful region depends on both the channel and the chosen structural operation. Our aim is to explain that dependence under a stated receiver model, rather than claim a universal threshold or an optimal estimator.

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II. System Model

We consider an uplink system with K single-antenna users and one Rydberg atomic receiver, as shown in Fig. 1. The receiver contains N vapor-cell receive elements arranged in a uniform linear array (ULA). Each vapor cell contains atoms that are excited to Rydberg states using probe and coupling lasers. The RF signals transmitted by the users interact with these atoms. A known RF reference field is also applied at the receiver. The atomic response changes the probe laser passing through the vapour cell, and this change is measured using a photodetector. The receiver therefore measures the strength of the total RF field, but it does not directly observe its complex phase [1].

We model the channel between each user and the array using several propagation paths. A path may represent a signal arriving directly from the user or after interacting with objects in the environment. For user k, the channel coefficient at the nth receive element is

$$g_k[n] = \sum_{\ell=1}^{L_k} \alpha_{\ell,k} e^{-\jmath(n-1)\psi_{\ell,k}}, \qquad \psi_{\ell,k} = \pi \sin\theta_{\ell,k}, \tag{1}$$

where L_k is the number of propagation paths for user k, $\alpha_{\ell,k}$ is the complex gain of path ℓ, and $\theta_{\ell,k}$ is its angle of arrival. Unless otherwise stated, L_k is drawn independently and uniformly from $\{3,\dots,7\}$, following the path-count setting in [4]. Fixed path counts are used in the complexity studies.

The exponential term in (1) describes how the phase of one propagation path changes as we move from one array element to the next. To see this, consider a single path arriving at the array from an angle θ. The signal reaches the different array elements at slightly different times because the elements are at different positions. This difference in travel distance produces a phase difference between adjacent elements.

With half-wavelength spacing between the array elements, the phase change between two neighbouring elements is

$$\psi = \pi \sin\theta. \tag{2}$$

The first array element has no additional phase shift from the array position. The second element has a phase shift of ψ, the third has a phase shift of 2ψ, and so on. Therefore, the samples produced by one path across the array have the form

$$1, \quad e^{-\jmath\psi}, \quad e^{-\jmath 2\psi}, \quad \dots, \quad e^{-\jmath(N-1)\psi}. \tag{3}$$

The factor

$$e^{-\jmath(n-1)\psi_{\ell,k}} \tag{4}$$

in (1) represents exactly this phase pattern for path ℓ. The coefficient $\alpha_{\ell,k}$ determines the strength and initial phase of that path. Since several paths can arrive at the receiver, their contributions are added together to produce the final channel coefficient $g_k[n]$.

The channel coefficients of user k are collected into the vector

$$\mathbf{g}_k = \begin{bmatrix} g_k[1] & g_k[2] & \cdots & g_k[N] \end{bmatrix}^T, \tag{5}$$

and the channels of all K users are collected into

$$\mathbf{G} = \begin{bmatrix} \mathbf{g}_1 & \mathbf{g}_2 & \cdots & \mathbf{g}_K \end{bmatrix} \in \mathbb{C}^{N\times K}. \tag{6}$$

During channel estimation, the K users transmit P known pilot symbols. These pilots are collected in

$$\mathbf{S} \in \mathbb{C}^{K\times P}. \tag{7}$$

Each row of $\mathbf{S}$ contains the pilots transmitted by one user, while each column represents one pilot transmission. The same pilot matrix is used when comparing the different channel estimators.

If the receiver could observe the complete complex signal, the contribution from the users would be $\mathbf{GS}$. However, the Rydberg receiver considered here measures magnitude rather than directly measuring this complex value. A known reference RF field is therefore applied at the receiver. We denote this reference by

$$\mathbf{B} \in \mathbb{C}^{N\times P}. \tag{8}$$

The reference field is known in both amplitude and phase. It is important because the magnitude of an unknown signal alone does not directly reveal its phase. When the unknown user signal is combined with a known reference, the resulting magnitude depends on their relative phase. This provides information that can be used to estimate the complex channel.

Including noise, the field before the magnitude measurement is

$$\mathbf{Y} = \mathbf{GS} + \mathbf{B} + \mathbf{W}, \tag{9}$$

where the noise is modeled as complex Gaussian,

$$\mathbf{W}_{np} \sim \mathcal{CN}(0,\sigma^2). \tag{10}$$

The noise is therefore added before the magnitude is measured. The Rydberg receiver observes

$$\mathbf{Z} = \left|\mathbf{GS} + \mathbf{B} + \mathbf{W}\right|, \tag{11}$$

where $|\cdot|$ is applied to every element of the matrix.

Equation (11) is the main measurement model used in this work. The pilots $\mathbf{S}$ and reference field $\mathbf{B}$ are known, and the receiver measures $\mathbf{Z}$. The unknown quantity is the complex channel matrix $\mathbf{G}$. The channel-estimation problem is therefore to recover $\mathbf{G}$ even though the receiver does not directly observe the phase of the received signal.

We assume a narrowband, frequency-flat channel during the pilot interval. The path gains are modeled as independent circularly symmetric complex Gaussian variables, and the angles of arrival are drawn uniformly over $(-90^\circ, 90^\circ)$, subject to the numerical conditioning checks stated in Section V. The users have equal statistical received power, and the reference field remains known at the receiver. For a fair comparison, all estimators use the same channel, pilots, reference field, and noise realization within each Monte Carlo trial.

The reference-to-signal ratio (RSR) is the reference power divided by the single-user received signal power. The signal-to-noise ratio (SNR) is the total received signal power from

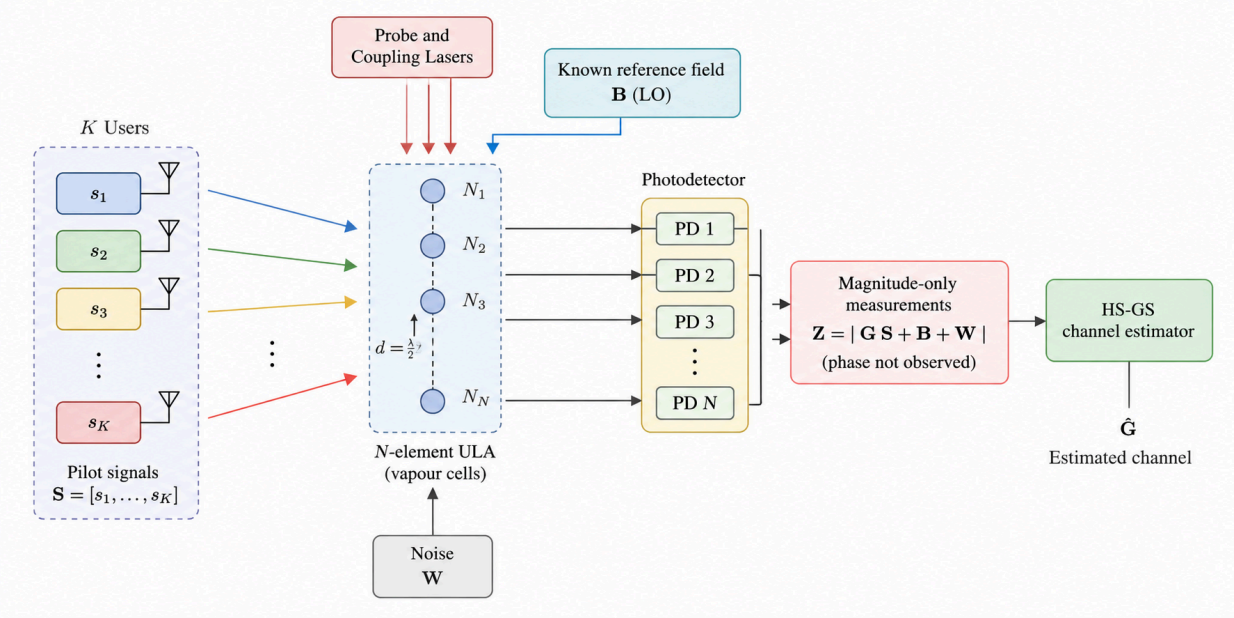


Fig. 1. Uplink of the Rydberg atomic receiver. K single-antenna users transmit known pilot signals to N vapour-cell receive elements arranged as a uniform linear array. The user RF signals and a known reference RF field interact with the Rydberg atoms. Probe and coupling lasers are used for the atomic readout, and the photodetector measures the magnitude of the total field. The estimator therefore observes $\mathbf{Z}$ without directly observing its phase.

all users divided by σ^2. We use this aggregate SNR definition throughout the simulation results.

III. Proposed Estimator

A. Structure and its Capacity

The rank of a matrix tells us how many independent components are needed to describe it. A low-rank matrix is therefore described using only a small number of important components. To use this idea for our channel, we first arrange the channel into a Hankel matrix. A Hankel matrix is a matrix in which the same channel samples are repeated along the antidiagonals. For example, consider

$$\mathbf{g} = \begin{bmatrix} g_1 & g_2 & g_3 & g_4 & g_5 \end{bmatrix}^T. \tag{12}$$

One possible Hankel representation is

$$\mathcal{H}(\mathbf{g}) = \begin{bmatrix} g_1 & g_2 & g_3 \\ g_2 & g_3 & g_4 \\ g_3 & g_4 & g_5 \end{bmatrix}. \tag{13}$$

We use this representation because the channel in (1) is not a random collection of values. It is made up of propagation paths, and each path produces a complex exponential pattern across the array elements. When this type of channel is written as a Hankel matrix, each path contributes one rank-one component. For example, a channel containing one path produces a Hankel matrix with rank at most one. Two paths produce a matrix with rank at most two. In general, for a channel with L_k paths,

$$\operatorname{rank} \mathcal{H}_p(\mathbf{g}_k) \leq L_k. \tag{14}$$

Therefore, when the channel has only a few important paths, its Hankel matrix can have a low rank [7], [8]. This is why we use the Hankel representation: it changes the multipath structure of the channel into a low-rank matrix structure that can be used during channel estimation.

The maximum possible rank of the Hankel matrix depends on the number of array elements. For an $(N-p) \times (p+1)$ Hankel matrix, the maximum rank is

$$\min(N-p, p+1). \tag{15}$$

Choosing p so that the matrix is as balanced as possible gives

$$r_{\max}(N) = \left\lceil \frac{N}{2} \right\rceil. \tag{16}$$

For example, if $N = 16$, the maximum Hankel rank is 8. If the channel has only three important paths, its Hankel rank can be at most 3. This is much smaller than the maximum rank of 8, so there is useful low-rank structure that can be used during estimation. If the channel rank is already close to 8, there is less low-rank structure available. In this case, forcing the channel to have a much smaller rank may remove part of the real channel.

B. The Estimator

HS-GS uses the same measurement update and spectral initialization as EM-GS. It then applies the Hankel step separately to each user column $\mathbf{g}_k$. We do not stack different users or pilot measurements into one Hankel matrix, and we do not assume that users share arrival angles.

At iteration t, the measurement update is

$$\mathbf{M}^{(t)} = \mathbf{G}^{(t)}\mathbf{S} + \mathbf{B}, \tag{17}$$

$$\boldsymbol{\kappa}^{(t)} = 2\mathbf{Z} \odot |\mathbf{M}^{(t)}|/\sigma^2, \tag{18}$$

$$\mathbf{Y}^{(t)} = \mathbf{Z} \odot R(\boldsymbol{\kappa}^{(t)}) \odot e^{\jmath \arg \mathbf{M}^{(t)}}, \tag{19}$$

$$\widetilde{\mathbf{G}}^{(t+1)}(\mathbf{S}\mathbf{S}^H) = (\mathbf{Y}^{(t)} - \mathbf{B})\mathbf{S}^H, \tag{20}$$

where $R(x) = I_1(x)/I_0(x)$ and all nonlinear operations are elementwise. The last equation is solved as a linear system. The pilot matrix for each fit is assumed to have full row rank. Scaled Bessel functions are used to evaluate the ratio stably; the contribution to $\mathbf{Y}^{(t)}$ is zero when the corresponding entry of $\mathbf{M}^{(t)}$ is zero.

For a channel vector of length N, we use

$$p = \left\lfloor \frac{N-1}{2} \right\rfloor, \qquad [\mathcal{H}_p(\mathbf{g})]_{a+1,b+1} = g[a+b+1], \tag{21}$$

with $0 \le a < N-p$ and $0 \le b \le p$. This gives an $(N-p) \times (p+1)$ matrix and the exact capacity $r_{\max} = \lceil N/2 \rceil$.

One Cadzow sweep is

$$\mathcal{P}_r(\mathbf{g}) = \mathcal{H}_p^{\dagger}(\Pi_r(\mathcal{H}_p(\mathbf{g}))). \tag{22}$$

Here Π_r keeps the r largest singular values and sets the others to zero. It does not change the retained singular values. The operation $\mathcal{H}_p^{\dagger}$ averages the entries on each antidiagonal to recover the channel vector. We apply four sweeps to every column of $\widetilde{\mathbf{G}}^{(t+1)}$ before the next measurement update.

This is a finite sequence of structural corrections. After antidiagonal averaging, the Hankel matrix formed again from the output can have rank greater than r. Low Hankel rank also does not impose every physical constraint in the geometric channel model. A small rank can remove useful channel components, while a large rank may leave much of the estimation error unchanged.

C. Choosing the Hankel Rank

We choose the rank from the available pilots without using the true path count. The last $P_B = \max\{1, \mathrm{round}(0.3P)\}$ pilot columns form the validation set B. The remaining columns form the fitting set A. For example, with $P = 30$, we use 21 columns to fit each candidate and nine columns to compare the candidates.

For every $r \in \{1, \ldots, r_{\max}\}$, we run the estimator on set A and obtain $\widehat{\mathbf{G}}_A^{(r)}$. Its initialization also uses only set A. We then select

$$\widehat{r} = \underset{1 \le r \le r_{\max}}{\arg\min} \left\| \mathbf{Z}_B - \left| \widehat{\mathbf{G}}_A^{(r)} \mathbf{S}_B + \mathbf{B}_B \right| \right\|_F^2. \tag{23}$$

The score sums the squared magnitude residuals over all receive elements and validation columns. If scores are exactly tied, we select the smallest rank. One scalar rank is shared by all user columns; the Hankel operations on those columns remain separate.

After choosing the rank, we restart the estimator from an initialization computed using all P pilots and fit the final estimate using all of them. Thus, the validation split does not permanently discard 30% of the pilots or require additional pilot transmissions. Its costs are the repeated candidate fits and the smaller fitting set used during rank selection.

If $\widehat{r} = r_{\max}$, the structural step is bypassed. With the same initialization and update count, the final estimate then equals the paired EM-GS estimate exactly. This fallback does not guarantee that a smaller selected rank improves the unknown channel error in every trial.

HS-GS procedure
Inputs: $\mathbf{Z}, \mathbf{S}, \mathbf{B}, \sigma^2$; candidate and final update counts T_{sel}, T; four Cadzow sweeps per application.

1) Form fitting set A and validation set B from the pilot columns.
2) For each candidate rank, initialize from A and perform T_{sel} measurement updates, each followed by four per-user Cadzow sweeps. Bypass these sweeps at $r_{\max}$.
3) Select $\widehat{r}$ using (23).
4) Reinitialize using all pilots. Perform T measurement updates with the selected structural processing and return the final channel estimate.

D. Computational Cost

Let R be the number of candidate ranks and Q the number of Cadzow sweeps per application. One EM-GS update costs $O(NKP)$ after the pilot least-squares operator has been formed. A balanced Hankel matrix has both dimensions proportional to N. Using a full SVD, one sweep costs $O(KN^3)$ across the K users; antidiagonal averaging costs $O(KN^2)$. Selecting a small rank does not reduce this full-SVD cost.

With T_{sel} candidate updates and T final updates, an upper order bound for the iterative work is

$$O\big((RT_{\mathrm{sel}} + T)(NKP + QKN^3)\big). \tag{24}$$

This uses P as an upper bound on the fitting-set size. Full-rank candidates bypass structural processing and cost less. The spectral initializer adds $O(NPK^2 + NK^3)$ work per fitting set. Forming the pilot operator adds $O(PK^2 + K^3)$. Our candidate fits share the same initializer and pilot operator; both are recomputed for the all-pilot final fit.

The implementation batches candidates and users. Its iterative storage scales as $O(R(NP + NK + KN^2) + PK + K^2)$, including the Hankel/SVD arrays. The initializer additionally stores per-row diagnostics of order $O(NPK + NK^2)$. Processing candidates or users sequentially can reduce peak storage without changing the estimator. These expressions describe scaling; measured times also depend on the implementation and hardware.

IV. Performance References

A. Information in an Amplitude Measurement

The Cramér–Rao lower bound (CRLB) is used as a theoretical reference for channel estimation. In simple terms, it tells us how accurately an unbiased estimator could estimate the channel from the available measurements. The amount of information contained in the measurements therefore affects the CRLB.

This is important for the Rydberg receiver because it measures only the amplitude of the received signal, not its full

complex value. For example, suppose the noiseless received signal is

$$s = 3 + j4. \tag{25}$$

A complex receiver can observe both the real part, 3, and the imaginary part, 4. The amplitude-only receiver instead observes

$$|s| = \sqrt{3^2 + 4^2} = 5. \tag{26}$$

Knowing only the value 5 does not tell us that the original signal was $3 + j4$. Other complex values can have the same amplitude. Therefore, taking magnitude can discard information present in a full complex measurement.

When complex Gaussian noise is added before the amplitude is measured, the measured amplitude follows a Rician distribution. We use this Rician model to calculate the Fisher information matrix J and the corresponding CRLB [13], [1]. The Fisher information matrix describes how much information the measurements contain about the unknown channel.

For $s = x+jy$, the amplitude is $a = \sqrt{x^2 + y^2}$. At $(x, y) = (3, 4)$, its gradient is

$$\nabla a = \begin{bmatrix} 3/5 & 4/5 \end{bmatrix}^T. \tag{27}$$

This tells us the direction in which this amplitude measurement provides information about the unknown complex signal. The Fisher information from this measurement has the form

$$J \propto \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} \begin{bmatrix} 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{bmatrix}. \tag{28}$$

This matrix has rank one. This means that one amplitude measurement gives information in only one direction. We do not have enough information from this one measurement to find the real and imaginary parts separately. This agrees with the earlier example: knowing that the amplitude is 5 is not enough to know that the signal is exactly $3 + j4$.

Independent pilot measurements contribute information that adds to form the full Fisher information matrix. Their directions depend on both the pilots and the known reference. The spatial channel model provides additional information, which motivates the constrained reference below.

B. Constrained Cramér–Rao Bound

The ordinary CRLB treats the real and imaginary parts of each channel coefficient as separate unknowns. The geometric CCRB also uses the fact that these coefficients are generated by a common set of paths. It is

$$\mathbf{C}_{\text{geom}} = \mathbf{U}(\mathbf{U}^T\mathbf{J}\mathbf{U})^{-1}\mathbf{U}^T, \tag{29}$$

where $\mathbf{J}$ is the Fisher information matrix in real channel coordinates. The columns of $\mathbf{U}$ form an orthonormal basis for the local changes in the channel allowed by the geometric model [14], [15]. Appendix A gives the parameter ordering, derivatives and numerical calculation.

The bound is evaluated at the true channel and its path configuration. The path gains and spatial frequencies are still unknown coordinates in the calculation; they are not supplied to the estimator. The known quantities are the pilots, reference field and noise variance. The true model order is used only to define this ideal geometric reference.

We plot the trace of each bound with the same channel-energy normalization as the NMSE. The gap between the two references shows the potential value of the geometric information under their local unbiasedness assumptions. It does not prove that the gap is achievable by HS-GS or another practical estimator. EM-GS and HS-GS are not assumed to be unbiased, so neither curve is treated as an absolute lower limit on their measured NMSE.

V. Numerical Results

A. Experimental Setup

We use the exact magnitude model in (11). Unless otherwise stated, $K = 3$ and the path counts are drawn independently from three to seven. For each user, $\alpha_{\ell,k} \sim \mathcal{CN}(0, 1/L_k)$ and $\theta_{\ell,k}$ is drawn uniformly on $(-\pi/2, \pi/2)$. The pilots are drawn from $\mathcal{CN}(0, 1)$. These angle and pilot draws are subject to the numerical checks below. The known reference is a broadside field with constant waveform and zero phase. With the nominal single-user signal power normalized to one, its entries are $B_{np} = 10^{\mathrm{RSR_{dB}}/20}$, and $\sigma^2 = K10^{-\mathrm{SNR_{dB}}/10}$. Each real noise component has variance $\sigma^2/2$. These are statistical power normalizations; individual channel draws are not forced to have the same energy. The alternative-channel normalization is stated separately. Geometric angle draws are repeated if their minimum circular spatial-frequency separation is below 10^{-10} or the steering-matrix singular-value ratio $\sigma_{\min}/\sigma_{\max}$ is below 10^{-8}. Pilot matrices are also redrawn if this ratio is below 10^{-8}. These checks condition the simulated distributions and are part of the declared model used for the results.

The pilots have equal expected power but are not forced to be orthogonal. Their full row rank permits the EM least-squares solve; it does not by itself guarantee identifiable magnitude-only recovery. Pilot phases, conditioning and reference strength also affect the Fisher information. Optimizing the pilots or allocating unequal user powers would change the experimental design and is not studied here.

Every comparison uses the same channel, pilots, reference and noise for all estimators within a trial. All iterative methods use the same spectral initializer, recomputed on the available fitting pilots. HS-GS uses four Cadzow sweeps per structural application. Table I lists the main settings. The two update counts refer to the final fit and each candidate fit. The final fit always uses all P pilots.

The effective-rank study uses 400 trials per point for $N = 16, 32$ and 300 for $N = 64$. Random SNRs are drawn uniformly from $[-10, 20]$ dB and rounded to 0.001 dB. The user-count study pairs $(K, P) = (2, 13), (3, 20)$ and $(4, 27)$ with five paths per user. Main-performance trials are the $N = 32$, $P = 30$ subset of the array study, not an additional independent sample.

TABLE I
SIMULATION SETTINGS. UNLESS STATED OTHERWISE, $K = 3$. THE UPDATE COUNTS ARE FINAL/CANDIDATE UPDATES.

Experiment	N	P	RSR (dB)	SNR (dB)	Updates	Trials/point
Main performance	32	30	12	$\{-5, 0, 5, 10, 15, 20\}$	100/25	400
Array size	$\{8, 16, 32\}$	$\{10, 30\}$	12	$\{-5, 0, 5, 10, 15, 20\}$	100/25	400
Path count	32	30	12	5	50/20	300
Effective rank	$\{16, 32, 64\}$	20	10	$\mathcal{U}[-10, 20]$	100/25	300–400
Alternative channels	32	20	10	$\mathcal{U}[-10, 20]$	100/25	400
User/pilot count	32	$\{13, 20, 27\}$	10	$\mathcal{U}[-10, 20]$	100/25	400
Projection placement	32	30	12	$\{-5, 0, 5, 10, 15, 20\}$	100/25	300
Structured comparisons	32	30	12	$\{0, 10\}$	100/25	400
Validation fraction	32	$\{10, 30\}$	12	$\{0, 10\}$	100/25	400
Position errors	32	30	12	5	100/25	400
Sweep/split stability	32	30	12	$\{0, 10\}$	100/25	100
Reference strength	32	30	$\{0, 5, 10, 12, 15, 20, 25\}$	5	100/25	400

We report two error summaries. Let $e_{m,i} = \|\widehat{\mathbf{G}}_{m,i} - \mathbf{G}_i\|_F^2$ and $d_i = \|\mathbf{G}_i\|_F^2$ for method m in trial i. The plotted NMSE is

$$\mathrm{NMSE}_{m,\mathrm{dB}} = 10 \log_{10} \frac{\sum_i e_{m,i}}{\sum_i d_i}. \tag{30}$$

The paired-median HS-GS gain is

$$\Delta_{\mathrm{med}} = \mathrm{median}_i \left[10 \log_{10} \frac{e_{\mathrm{EM},i}}{e_{\mathrm{HS},i}} \right]. \tag{31}$$

The first summary gives more weight to trials with large squared errors. The second describes a typical paired improvement. The vertical difference between NMSE curves therefore need not equal the paired median. Reported 95% intervals use 4,000 paired bootstrap resamples. They are pointwise intervals, not simultaneous guarantees over all conditions. Averages over SNR use equal weight for each stated operating point.

The main verification campaign contains 27,300 paired trials across 73 settings, including auxiliary diagnostics, and 2,400 bound evaluations on the main-figure channels. The added baseline, split and position-error studies contain 12,000 condition-trial evaluations from 4,800 independent base realizations. Split fractions reuse the same observations, while position-error strengths reuse the underlying paths, pilots and noise. The separate update-budget check uses 300 further trials. Reference-strength results reuse 2,800 trials from a separate campaign, and the rank-proxy analysis replays EM-GS on 10,400 existing main-campaign trials. Replays and paired repeats are not counted as new independent channels. Trial counts and candidate grids are fixed before inspecting the corresponding results. The sweep-count and random-split checks in Appendix C add 600 independent worlds, reused across all their variants, and a separate 12-world serial timing check.

B. Performance Against the Reference Curves

We first compare EM-GS and HS-GS with the CRLB and CCRB discussed in Section IV. Fig. 2(a) shows the NMSE as the aggregate SNR changes from -5 to 20 dB. This experiment uses $N = 32$, $K = 3$, $P = 30$, and an RSR of 12 dB.

HS-GS performs better than EM-GS at all six SNR values. The median improvement across the paired trials ranges from 2.54 to 2.97 dB. The average of these six median improvements is 2.70 dB. The vertical distance between the two NMSE curves ranges from 2.38 to 2.90 dB. These values are slightly different because the paired median and the NMSE curves summarize the trial errors in different ways.

For this experiment, the selected Hankel rank is smaller than the maximum rank in all 2,400 trials. Therefore, the Hankel projection is active in every trial.

At SNR values of 5 dB and above, the EM-GS result differs from the ordinary Rician CRLB by less than 0.108 dB. This shows that the ordinary CRLB is a useful reference for the EM-GS baseline.

The geometric CCRB is approximately 6.95–7.06 dB below the ordinary CRLB. It includes the local geometric multipath structure, while the ordinary CRLB treats the channel coefficients as separate unknowns. HS-GS improves on EM-GS but remains 4.37–4.85 dB above the geometric reference. These gaps describe the two reference curves and the measured estimator performance under this model. They do not establish how much further improvement a practical estimator can achieve.

The CRLB and CCRB are reference curves for unbiased estimators. EM-GS and HS-GS are not assumed to be unbiased, so these curves are not treated as absolute lower limits on their measured NMSE.

We define the vertical gain between the NMSE curves as

$$\Delta_{\mathrm{HS}} = \mathrm{NMSE}_{\mathrm{EM-GS,dB}} - \mathrm{NMSE}_{\mathrm{HS-GS,dB}}. \tag{32}$$

A positive value of Δ_{HS} means that HS-GS has a lower NMSE than EM-GS.

C. Effect of Array Size

We next study how the number of receiver elements affects the benefit of the Hankel constraint. We test

$$N \in \{8, 16, 32\}. \tag{33}$$

For each array size, we use two pilot counts, $P = 10$ and $P = 30$, and six SNR values. Each row in Table II therefore summarizes 12 operating points.

Fig. 2(b) and Table II show that the benefit of HS-GS increases with the array size. The mean of the 12 per-setting

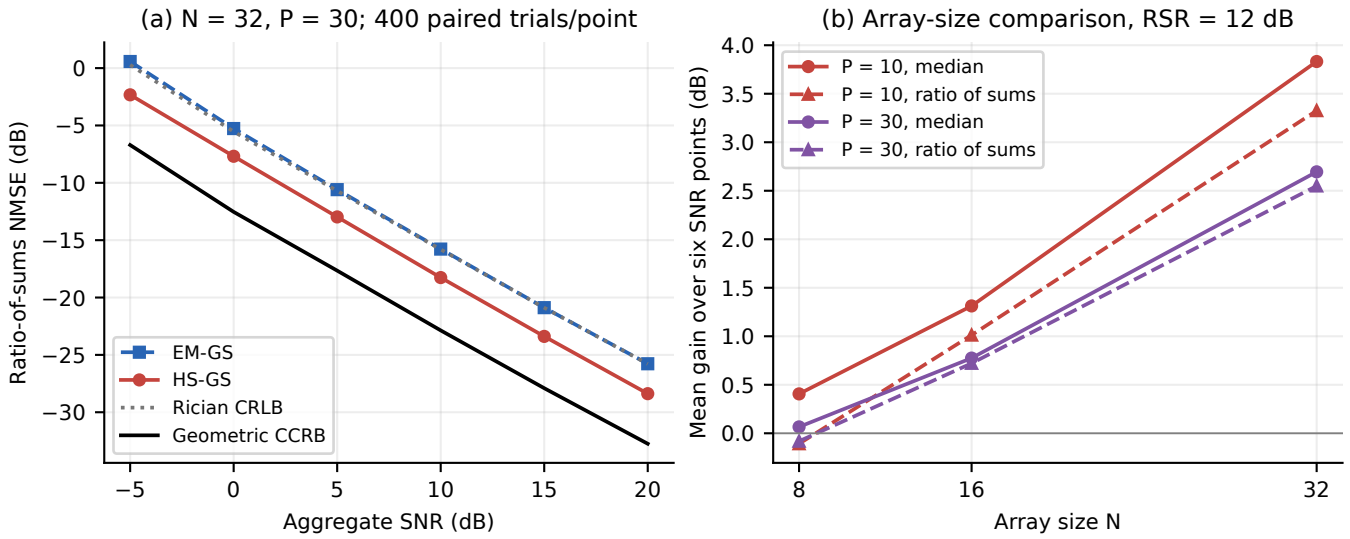


Fig. 2. Performance of EM-GS and HS-GS. (a) Ratio-of-sums NMSE versus aggregate SNR for $N = 32$, $P = 30$ and RSR $= 12$ dB, with the Rician CRLB and geometric CCRB. Each SNR uses 400 paired trials. (b) HS-GS gain versus array size for $P = 10$ and $P = 30$, averaged over six SNR values. Both the ratio-of-sums and paired-median summaries are shown.

TABLE II
EFFECT OF ARRAY SIZE, AVERAGED OVER 12 OPERATING POINTS. ALL VALUES EXCEPT THE ARRAY SIZE ARE IN DB.

N	Ratio-of-sums gain	Paired median	EM-GS NMSE
8	−0.095	0.236	−8.666
16	0.866	1.044	−8.786
32	2.938	3.264	−8.733

paired-median gains increases from 0.236 dB at $N = 8$ to 3.264 dB at $N = 32$.

This behavior can be understood from the maximum possible Hankel rank,

$$r_{\max} = \left\lceil \frac{N}{2} \right\rceil . \tag{34}$$

For the three tested array sizes,

N	8	16	32
$r_{\max}$	4	8	16

. (35)

When $N = 8$, the Hankel matrix can have a maximum rank of only four. Since the simulated channels normally contain between three and seven paths, there may be little unused rank available. A low-rank projection can then remove part of the actual channel instead of removing only noise and estimation error.

When $N = 32$, the maximum Hankel rank is 16. The same channel now occupies a smaller part of the available rank. This gives HS-GS more low-rank structure to use, which explains why its improvement is much larger.

The EM-GS baseline changes very little as the array size increases. Its average NMSE changes by only 0.120 dB between the tested array sizes. Without the Hankel constraint, increasing the array size adds more measurements, but it also adds more unknown channel coefficients.

At $N = 8$, the mean of the 12 paired-median gains is 0.236 dB, while the mean of the 12 ratio-of-sums gains is −0.095 dB. These are averages of operating-point summaries, not summaries obtained by pooling every $N = 8$ trial together. Within each operating point, large errors can have more influence on the ratio of sums than on the paired median.

The Hankel projection is inactive in 42.94% of the $N = 8$ trials. In these trials, HS-GS gives exactly the same estimate as EM-GS. An active projection can still increase channel error in other trials. The fallback rate therefore does not guarantee a positive gain, and the two aggregation rules should be reported separately.

The projection is inactive in only 6.38% of the $N = 16$ trials and 0.31% of the $N = 32$ trials. This provides further evidence that the Hankel constraint becomes more useful as the array provides more structural capacity.

D. Effect of Channel Complexity

We next study whether the number of propagation paths affects the benefit of the Hankel constraint. We keep the array size fixed at $N = 32$ and the SNR at 5 dB, and change only the number of paths L. The same value of L is used for all users.

The results show a clear trend. When the channel has only $L = 2$ paths, HS-GS improves over EM-GS by 7.377 dB. As the number of paths increases, the improvement becomes smaller. At $L = 16$, the gain becomes −0.125 dB. These are ratio-of-sums gains: at $L = 16$, the measured HS-GS NMSE is slightly higher than the EM-GS NMSE. Its paired-median gain is zero. This result concerns the tested hard-rank procedure.

This can be understood from the rank of the Hankel matrix. Since $N = 32$, the maximum Hankel rank is

$$r_{\max} = \left\lceil \frac{32}{2} \right\rceil = 16. \tag{36}$$

Consider the two ends of the experiment. With only two paths, the channel can have a Hankel rank of at most 2, while the matrix can support a rank of up to 16. The channel is therefore strongly low rank, and there is useful structure for HS-GS to use. In contrast, when $L = 16$, the channel can use most or all of the available Hankel rank. In this case, forcing a low-rank approximation can remove an important part of the real channel instead of only removing estimation error.

The EM-GS baseline changes by only 0.129 dB over the same experiment. Therefore, the large change in HS-GS gain mainly comes from the changing usefulness of the Hankel constraint.

These results show that HS-GS works best when the channel has relatively few important components compared with the maximum Hankel rank. However, simply counting the number of paths is not always enough to describe the real complexity of the channel. We take a closer look at this next using effective rank.

E. Effective Rank

Two channels can have the same number of paths but still have different levels of complexity. For example, consider two channels that both have four paths. In the first channel, all four paths may have similar strength. In the second channel, two paths may be strong while the other two are very weak. Although both channels have $L = 4$, the second channel can behave more like a channel with only two important components.

The singular values of the Hankel matrix help us measure this difference. A large singular value represents an important component of the matrix, while a very small singular value represents a weak component. For example, consider

$$[10,\ 9,\ 8,\ 7] \tag{37}$$

and

$$[10,\ 8,\ 0.2,\ 0.1]. \tag{38}$$

Both matrices can have rank 4. However, in the second case, most of the matrix can be explained by the first two components. Its real complexity is therefore much smaller than its ordinary rank suggests.

To measure this, we use the effective rank proposed in [16],

$$r_{\text{eff}} = \exp\left(-\sum_i p_i \ln p_i\right), \qquad p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \tag{39}$$

where σ_i are the singular values of the Hankel matrix. The value p_i represents the relative importance of each singular value. If many singular values are important, r_{eff} is large. If only a few are important, r_{eff} is small.

We then compare the effective rank with the highest rank that the Hankel matrix can support. We define

$$\rho = \frac{r_{\text{eff}}}{r_{\max}}. \tag{40}$$

We calculate this quantity separately for each user column. At an operating point, the reported ρ is the median over the user columns and channel realizations. When resampling trials, we keep all users from the same realization together. The values shown here are calculated from the true noiseless channels and are used to explain the simulation results; they are not inputs to rank selection.

The ratio ρ describes how much of the available Hankel rank the channel effectively uses. For example, $r_{\text{eff}} = 4$ and $r_{\max} = 16$ give $\rho = 0.25$, whereas $r_{\text{eff}} = 14$ gives $\rho = 0.875$. The second channel leaves less room for a low-rank approximation. This is an interpretation of the noiseless channel, not a guarantee about the estimate.

Fig. 3 shows the HS-GS gain against ρ for $N \in \{16, 32, 64\}$. The same pilot count, RSR, iteration settings, SNR distribution, and performance measure are used for all three array sizes. The tested path counts are $L \in \{2, 4, 7, 9, 14\}$ for $N = 16$, $L \in \{2, 4, \ldots, 16\}$ for $N = 32$, and $L \in \{8, 14, 29, 38, 48\}$ for $N = 64$. The plotted gain is the pooled paired median over the stated random-SNR distribution. It is distinct from the ratio-of-sums gain used for the NMSE curves.

The results show that the HS-GS gain becomes smaller as ρ increases. When ρ is small, the channel uses only a small part of the available Hankel rank and the hard-rank processing used by HS-GS provides a clear improvement. When ρ approaches approximately 0.5, the improvement becomes small.

For example, at $N = 32$ and $\rho = 0.536$, the median gain is 0.141 dB, with a 95% confidence interval of $[0.090, 0.211]$ dB. The gain is still positive, but it is small in practice. Therefore, $\rho \approx 0.5$ should be understood as a rough description of the region where the tested HS-GS becomes less useful, rather than as an exact threshold.

We also test what happens when a low-rank constraint is forced on a more complex channel. For $N = 16$ and $L = 14$, the measured normalized effective rank is

$$\rho = 0.732. \tag{41}$$

When rank seven is forced, the median HS-GS gain becomes -0.154 dB, with a 95% confidence interval of $[-0.201, -0.111]$ dB. The negative gain shows that forcing a low-rank constraint on a complex channel can make the estimate worse.

With adaptive rank selection, the pooled median gain is zero for the same channels. This happens because the estimator can select the maximum rank and return to EM-GS when a useful lower-rank constraint is not identified. However, the rank-selection procedure does not guarantee an improvement in every individual trial.

These results suggest that ρ is a useful way to describe when the tested HS-GS helps. However, ρ is calculated here using the true noiseless channel, which is not known at the receiver. We therefore use ρ to understand the simulation results rather than as a value used directly by the estimator.

We also compare effective rank with a simpler description of complexity, $L/r_{\max}$. In a separate prediction check, we fit a piecewise-linear gain predictor to the eight matched $N = 32$ operating points and evaluate it on six points: $N = 16$, $L \in \{2, 4, 7\}$ and $N = 64$, $L \in \{8, 14, 29\}$. The mean ab-

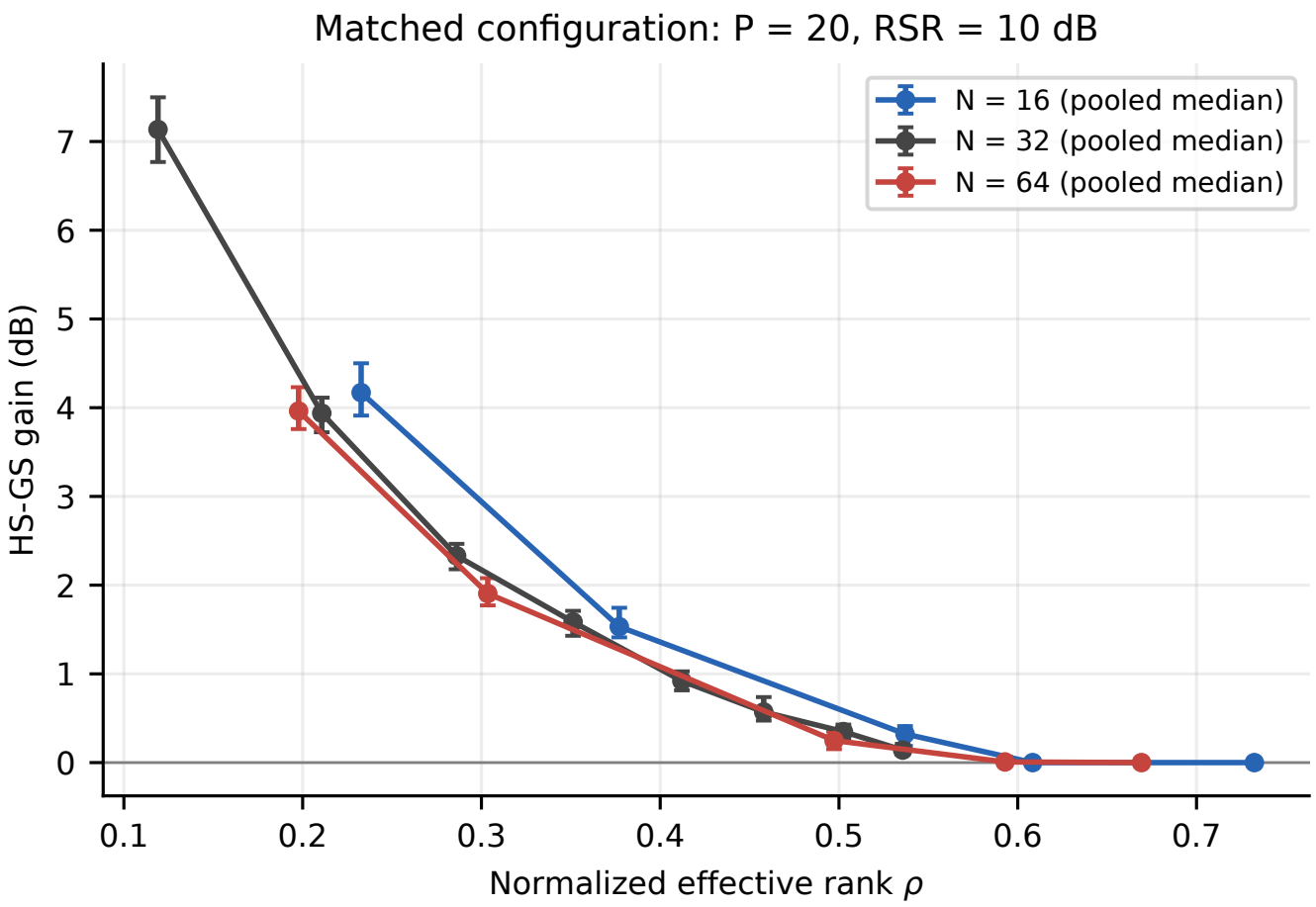


Fig. 3. Pooled paired-median HS-GS gain against the normalized effective rank $\rho = r_{\text{eff}}/r_{\max}$ for $N \in \{16, 32, 64\}$. The same simulation settings and performance measure are used for all three array sizes. Error bars show pointwise 95% intervals. The HS-GS gain becomes small near half of the available rank in this tested design; this is not a universal boundary for structural processing.

solute prediction errors are 2.067 dB using L, 0.073 dB using $L/r_{\max}$, and 0.317 dB using ρ. The 95% bootstrap interval for the error of the ρ predictor minus that of the $L/r_{\max}$ predictor is $[0.086, 0.283]$ dB. Thus, normalized path count is the better predictor in this check. This comparison is limited to the stated operating points and interpolation rule, with some endpoint extrapolation. All six target gains are positive, so it does not test the ability to detect harmful structural processing. Both quantities also use channel information that is unavailable at the receiver. We therefore treat ρ as a useful description of the results, without claiming that it is the best or a receiver-ready predictor of gain.

F. Additional Results

We test $(K, P) = (2, 13), (3, 20), (4, 27)$ with $N = 32$, five paths per user, RSR $= 10$ dB and the random SNR distribution in Table I. Among the trials with SNR at least 5 dB, the paired-median HS-GS gains are 3.381, 3.509 and 3.552 dB. Their 0.170 dB span describes this joint change in users and pilots, with approximately fixed $P/(2K)$. It does not isolate the effect of user count alone.

We also test two different path distributions at $N = 32$ and $P = 20$. The clustered channel contains four clusters with ten rays each. Cluster centres are uniform over $(-\pi/2, \pi/2)$, with ray offsets uniform within $\pm 5^\circ$ and angles clipped to the same range. The second channel has ten independently drawn paths. The path gains are independent complex Gaussian variables with variance equal to the reciprocal of the total path count. In each trial, the resulting channel is rescaled to the Frobenius norm of its paired baseline channel. RSR, random SNR and update counts follow Table I.

The paired-median gains are 1.182 dB for the clustered channel, with 95% interval $[0.995, 1.398]$ dB, and 0.831 dB for the ten-path channel, with interval $[0.710, 1.011]$ dB. These positive gains show that HS-GS remains useful for these two path distributions. Both retain the geometric ULA model, so this test alone does not establish robustness to array or propagation-model errors.

G. Projection Placement and Computation

We compare Hankel processing after every EM-GS update with processing only once after EM-GS has finished. The two variants use the same channel, pilots, reference, noise and selected rank. There are 300 paired trials at each of six SNRs, and four Cadzow sweeps per structural application. The shared

rank is selected using four-sweep interleaved candidates. This isolates placement conditional on that rank; the variants are not separately tuned. Equal sweeps per application also do not mean equal total computation.

The mean of the six per-SNR paired-median interleaving gains is 0.204 dB, with a paired-bootstrap interval of $[0.169, 0.242]$ dB. Interleaving therefore gives a small positive improvement in this design. This does not imply that it is always preferable under a fixed runtime budget or that the two estimates are otherwise identical.

A separate placement timing study uses $P = 20$, SNR $= 5$ dB, RSR $= 10$ dB, 100 final updates, 25 candidate updates and four sweeps. It measures three independent trials per array size on an Apple M2 with eight physical cores and 8 GiB RAM, using single-thread settings. At $N = 32$, median total times are 3.548 s for interleaving and 3.174 s for final-only processing. Rank selection accounts for 79.3% of the interleaved total at $N = 32$ and 88.5% at $N = 64$. These measurements describe that implementation and machine.

H. Comparison with Other Structural Processing

We compare six classical estimators: EM-GS, HS-GS, unstructured Rician gradient descent, Rician gradient descent with Hankel processing, EM-GS with smooth singular-value filtering, and the one-dimensional linearized least-squares objective of Xu et al. [4]. All methods receive the same exact magnitude observations. The linearized method is a comparison estimator; it does not change how the observations are generated. Both nonlinear data updates use the Rician likelihood. Omitting terms independent of $\mathbf{G}$, its negative log-likelihood is

$$\mathcal{L}(\mathbf{G}) = \sum_{n,p} \left[\frac{a_{np}^2}{\sigma^2} - \log I_0\left(\frac{2 Z_{np} a_{np}}{\sigma^2} \right) \right], \qquad (42)$$

where $a_{np} = |(\mathbf{GS} + \mathbf{B})_{np}|$. Scaled Bessel functions are used in the numerical calculation. Structural correction is not assumed to decrease this likelihood at every update. The Rician gradient method uses

$$\mathbf{G}_{\mathrm{dc}}^{(t+1)} = \mathbf{G}^{(t)} - \frac{(\mathbf{M}^{(t)} - \mathbf{Y}^{(t)})\mathbf{S}^H}{\|\mathbf{S}\|_2^2}, \qquad (43)$$

where $\mathbf{Y}^{(t)}$ is the Rician conditional field in Section III-B. The structured version follows this update with the same per-user Cadzow processing as HS-GS. It is an adapted Rician–Hankel control, rather than Xu et al.'s two-dimensional projected gradient method.

For the smooth-filtering control, each singular value s_i is replaced by

$$\widehat{s}_i = \frac{s_i^3}{s_i^2 + \tau^2}, \qquad \tau = f \sqrt{\frac{1}{q} \sum_{i=1}^{q} s_i^2}, \qquad (44)$$

where $q = r_{\max}$ and $f \in \{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$. The scale is recomputed from the current per-user Hankel spectrum in each of the four sweeps. A zero factor bypasses filtering. This is an empirical smooth filter, not OptShrink or a nuclear-norm solver. Each structured method selects its own rank or factor using the same held-out squared magnitude residual. The selected parameter is shared across users, and the final estimate is reinitialized and fitted using all pilots.

We choose this filter as a simple smooth alternative to hard truncation. For $s_i \gg \tau$, it changes the component little; for $s_i \ll \tau$, it suppresses the component strongly without an abrupt rank cutoff. The RMS scale makes the threshold follow the current singular-value amplitudes. The zero factor supplies an exact unstructured option. This choice tests whether soft suppression is useful; it is not derived as an optimal denoiser for the correlated errors produced by Hankel lifting.

We use $N = 32$, $K = 3$, $P = 30$, RSR $= 12$ dB, $L \in \{3, 7, 14\}$ paths per user and SNRs of 0 and 10 dB. Each of the six settings uses 400 paired trials. The iterative methods use 100 final updates and 25 updates per candidate, with four sweeps per structural application. These are fixed update budgets, not convergence guarantees. The gains below compare ratio-of-sums NMSE; intervals are pointwise 95% paired-bootstrap intervals from 4,000 resamples.

Fig. 4 shows that HS-GS is not the best method in every setting. Smooth filtering improves on HS-GS by 0.623–1.413 dB in five settings, with the interval above zero in each. The remaining case, $L = 3$ and SNR $= 10$ dB, gives a gain of 0.005 dB with interval $[-0.180, 0.191]$ dB. Rician–Hankel processing improves on HS-GS by 0.171 dB $[0.078, 0.274]$ at $L = 14$ and SNR $= 0$ dB, but is worse by 0.284 and 0.222 dB at SNR $= 10$ dB for $L = 3$ and $L = 7$, respectively. The two unstructured nonlinear methods have similar NMSE in these tests. The linearized least-squares method is less accurate than EM-GS at all six settings under this reference strength.

An important distinction appears at $L = 14$ and SNR $= 0$ dB. HS-GS has a gain of -0.147 dB $[-0.251, -0.046]$ over EM-GS, while smooth filtering has a gain of 1.266 dB $[1.152, 1.380]$. Thus, a loss for HS-GS does not show that every use of Hankel structure is unhelpful. The observed benefit also depends on how the structure is applied and how its parameter is selected.

We check the final update budget using 300 additional paired trials, with 50 trials for each combination of $L \in \{3, 7, 14\}$ and SNR $\in \{0, 10\}$ dB. The remaining settings are unchanged. For each method, the rank or shrinkage parameter is selected once using 25 candidate updates, and the same final trajectory is recorded at 100 and 300 updates. Across the five iterative methods and six operating points, the absolute change in ratio-of-sums NMSE is less than 0.06 dB. This check supports the chosen final update budget in the tested conditions, but it does not establish equivalence, global convergence, or the best possible rank-selection budget. Some estimates are still changing slightly after 300 updates. Parameters are not selected again using the longer trajectories or the true channel errors.

I. Sensitivity to Array Position Errors

To test one departure from the ideal ULA model, we perturb the element positions according to

$$\frac{x_n}{\lambda} = \frac{n-1}{2} + s_x u_n, \qquad u_n \sim \mathcal{N}(0, 1), \qquad (45)$$

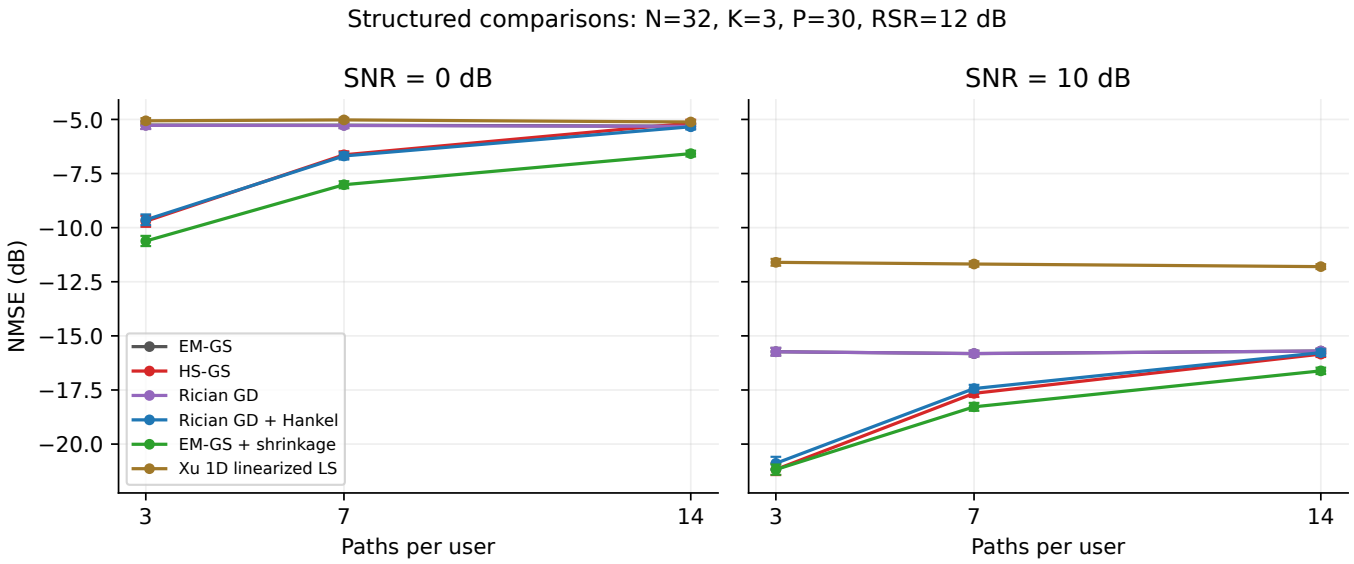


Fig. 4. Six classical estimators on the same exact magnitude observations. Each point uses 400 paired trials at $N = 32$, $K = 3$, $P = 30$ and RSR $= 12$ dB. The two panels use SNRs of 0 and 10 dB. Error bars are pointwise 95% paired-bootstrap intervals.

with independent u_n and $s_x \in \{0, 0.005, 0.01, 0.02\}$. These are declared stress levels, not measured hardware tolerances. The estimator continues to apply its ULA Hankel processing, and channel error is measured against the actual perturbed channel. The reference field remains known and calibrated.

We use $N = 32$, $K = 3$, $P = 30$, SNR $= 5$ dB, RSR $= 12$ dB and $L \in \{3, 12\}$ paths per user. Each setting uses 400 paired trials and the same 100/25 update counts, four sweeps and 30% validation fraction. For a given trial, the path parameters, pilots, thermal-noise draw and normalized offsets u_n are shared across the four strengths. We do not renormalize the realized channel after perturbing the positions; SNR retains its nominal statistical-power definition.

As shown in Fig. 5, the HS-GS gain decreases as the position errors increase. For $L = 3$, its ratio-of-sums gain over EM-GS changes from 4.955 dB without mismatch to 3.917 dB at $s_x = 0.02$, with interval $[3.748, 4.075]$ dB at the latter point. For $L = 12$, the gain decreases from 0.264 to 0.050 dB; the final interval, $[-0.029, 0.125]$ dB, includes zero. Thus, the tested sparse case retains a benefit, while the smaller benefit in the more complex case becomes uncertain. This experiment does not establish robustness to unknown reference calibration, mutual coupling, near-field propagation or correlated noise.

J. Runtime of the Comparison Methods

We also time the six classical methods in a separate serial experiment. This uses a vectorized implementation on the same machine, $N = 32$, $K = 3$, $P = 30$, $L = 7$, SNR $= 5$ dB and RSR $= 12$ dB. Table III reports medians over 20 fresh worlds with single-thread CPU execution and double precision. The iterative methods use 100 final updates, 25 updates per candidate and four structural sweeps. Timings include initialization, parameter selection where needed, and the final fit, but exclude channel and observation generation.

TABLE III
SERIAL RUNTIME OF THE CLASSICAL COMPARATOR IMPLEMENTATIONS.

Method	Median total time (ms)
EM-GS	9.943
HS-GS	348.536
Rician GD	10.015
Rician GD + Hankel	346.500
EM-GS + smooth filtering	200.799
Xu 1D linearized LS	0.647

The methods are timed in a fixed order for each world: HS-GS, Rician–Hankel, smooth filtering, EM-GS, Rician GD and linearized LS. No warm-up worlds are discarded. The median selection share is 76.0% for HS-GS and 56.4% for smooth filtering; the latter searches seven threshold factors rather than 16 ranks. These are measured costs for one implementation, machine and operating point. They are separate from the placement timings in Section V-G at $P = 20$ and RSR $= 10$ dB, so a difference between the two timing studies must not be interpreted as an algorithmic speedup.

K. Reference Strength, Pilot Splitting, and Rank Proxies

Further tests address practical choices behind HS-GS. At $N = 32$, $P = 30$ and SNR $= 5$ dB, the ratio-of-sums gain remains positive across the seven RSR values from 0 to 25 dB, ranging from 1.83 to 2.51 dB. A weak reference still worsens the absolute NMSE of both methods. Thus, a positive structural gain does not remove the difficulty caused by a weak reference. Appendix B gives the full reference sweep.

Validation fractions of 10%, 20%, 30% and 40% produce different ranks and errors. For example, at $P = 30$ and SNR $= 10$ dB, the 10% split is 0.602 dB worse than the 30% split. At $P = 10$ and the same SNR, the 40% split is 0.461 dB worse than 30%. All pilots are reused for the final estimate. The best

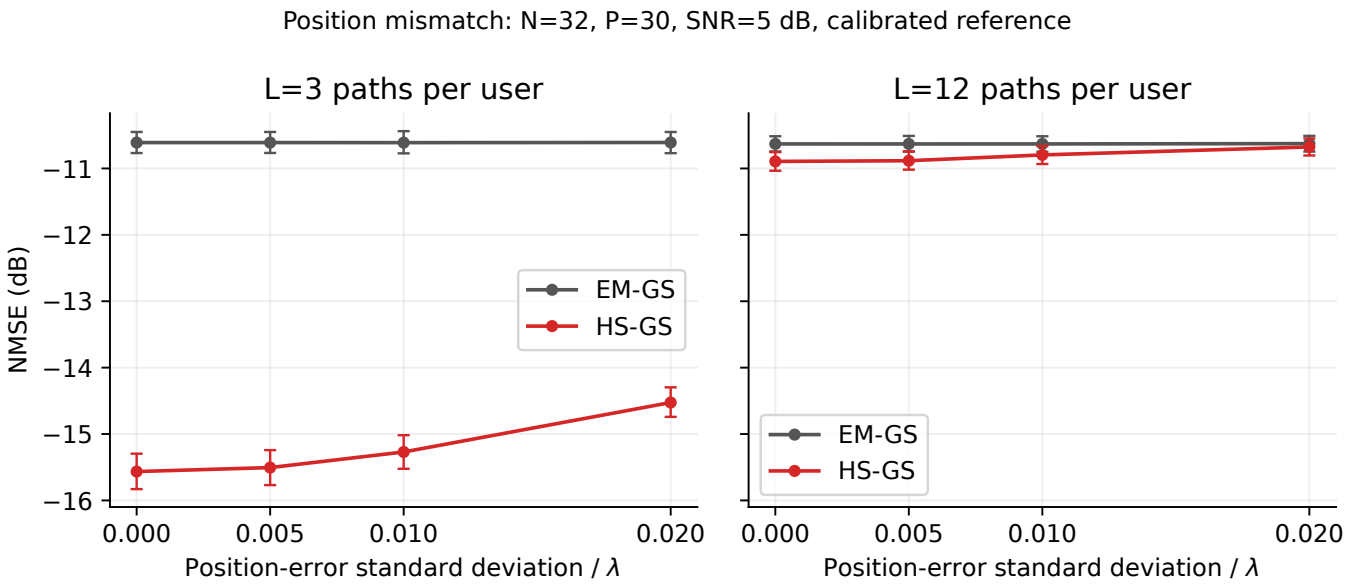


Fig. 5. Position-error stress test with a known calibrated reference and the actual perturbed channel as the estimation target. Each point uses 400 trials; trials are paired across mismatch strengths within each path count. Error bars are pointwise 95% paired-bootstrap intervals from 4,000 resamples.

fraction changes with the setting, so 30% is a declared choice, not a generally optimal one.

An effective-rank proxy computed from the unstructured EM-GS estimate tracks the true effective rank more closely at higher SNR. However, noise raises its value. At $N = 32$ and SNR $= 5$ dB, the mean proxy is 0.577 while the corresponding true-channel mean is 0.244, even though the paired-median HS-GS gain is 2.544 dB. The proxy study in Appendix B is descriptive; it does not validate a per-trial threshold or replace held-out rank selection.

VI. Limitations and Conclusion

This work studies how spatial Hankel processing affects magnitude-only channel estimation for the stated Rydberg receiver model. In the main experiment, HS-GS improves on EM-GS by an average of 2.70 dB across the six per-SNR paired medians. Its benefit increases with array capacity and generally decreases as channel complexity increases. Effective rank helps explain this trend, but it is not a universal predictor or a validated receiver-side decision rule.

The stronger comparisons change how this trend should be interpreted. With four sweeps per application, smooth singular-value filtering improves on hard-rank HS-GS in five of six comparison settings, by 0.623–1.413 dB with positive pointwise intervals. It can improve estimation even where HS-GS loses accuracy. Therefore, the point where the tested HS-GS becomes less useful is a property of its processing and rank-selection procedure, not a universal limit of Hankel structure. Hard truncation, the validation split and the update budget should all be treated as design choices.

Interleaving provides a small improvement over final-only processing under the shared-rank comparison, while rank selection remains the main measured computational cost. Reference strength and array-position errors also affect the benefit. The practical conclusion is to evaluate the structural rule and its parameter selection together for the intended channel and pilot conditions, including cases where the rule can cause harm. The focused sweep check reinforces this point: one hard-rank sweep outperforms the original four-sweep setting in all six tested conditions. Changing the held-out pilots often changes the selected rank, without a consistent average gain from random splitting. Appendix C reports these checks; the original curves retain their declared settings.

Limitations. All results are from simulations; we have not tested measured Rydberg receiver data. The additional path distributions retain the geometric channel model. The position-error study tests one departure from an ideal ULA with a known calibrated reference, but does not cover unknown calibration, mutual coupling, near-field propagation or correlated noise. Its offset levels are stress-test settings rather than hardware tolerances.

We process each user's spatial channel separately. Joint-user structure would require a model for shared paths, which the present independent-angle model does not provide. Likewise, the Gaussian pilot index is not assumed to have the exponential structure used by the spatial Hankel lift. Adaptive stopping and alternative parameter-selection rules remain open; this study does not establish convergence of the combined updates.

The effective rank of the true channel is unavailable at the receiver. The EM-GS proxy is informative at higher SNR but remains noise-sensitive; it does not establish a reliable per-trial switching rule.

The comparator results use the stated parameter grids and update budgets; they do not establish optimality over all structured estimators. The longer update check is limited to its tested conditions, and the timing results describe one implementation and machine. Finally, the CRLB and CCRB are local unbiased-estimation references. The tested estimators

need not be unbiased, so the bounds are not absolute lower limits on their NMSE.

APPENDIX A
REAL-COORDINATE FISHER INFORMATION AND GEOMETRIC CCRB

A. Channel coordinates and measurement information

For receive element n, define

$$\mathbf{x}_n = [\Re G_{n1}, \dots, \Re G_{nK}, \Im G_{n1}, \dots, \Im G_{nK}]^T. \tag{46}$$

The full real vector is $\mathbf{x} = [\mathbf{x}_1^T, \dots, \mathbf{x}_N^T]^T \in \mathbb{R}^{2NK}$. This ordering gives $\|\widehat{\mathbf{x}} - \mathbf{x}\|_2^2 = \|\widehat{\mathbf{G}} - \mathbf{G}\|_F^2$. We condition on the known $\mathbf{S}$, $\mathbf{B}$ and σ^2. The noise convention is $\mathbb{E}|W_{np}|^2 = \sigma^2$, so each real component has variance $\sigma^2/2$.

Write $\mu_{np} = (\mathbf{GS} + \mathbf{B})_{np}$ and $a_{np} = |\mu_{np}|$. For $z \geq 0$, the amplitude density is

$$p(z \mid a) = \frac{2z}{\sigma^2} \exp\left(-\frac{z^2 + a^2}{\sigma^2}\right) I_0\left(\frac{2za}{\sigma^2}\right). \tag{47}$$

With $R(u) = I_1(u)/I_0(u)$, its amplitude score is

$$\frac{\partial \log p(z \mid a)}{\partial a} = \frac{2}{\sigma^2}\left[zR\left(\frac{2za}{\sigma^2}\right) - a\right]. \tag{48}$$

Using $\mathbb{E}[zR(2za/\sigma^2)] = a$, the scalar information is

$$\mathcal{I}_a(a) = \frac{4}{\sigma^4}\left[\mathbb{E}\left\{z^2 R^2\left(\frac{2za}{\sigma^2}\right)\right\} - a^2\right], \tag{49}$$

$$\beta(a) = \mathcal{I}_a(a)/4. \tag{50}$$

For $\mu_{np} \neq 0$, put

$$\mathbf{c}_{np} = e^{-\jmath \arg \mu_{np}} \mathbf{S}_{:,p}, \qquad \mathbf{v}_{np} = \begin{bmatrix} \Re \mathbf{c}_{np} \\ -\Im \mathbf{c}_{np} \end{bmatrix}. \tag{51}$$

Here $\mathbf{v}_{np} = \partial a_{np}/\partial \mathbf{x}_n$. Independence of the measurement noises gives

$$\mathbf{J}_n = \sum_{p=1}^{P} \mathcal{I}_a(a_{np}) \mathbf{v}_{np} \mathbf{v}_{np}^T, \tag{52}$$

$$\mathbf{J} = \operatorname{blkdiag}(\mathbf{J}_1, \dots, \mathbf{J}_N). \tag{53}$$

Each nonzero measurement term has rank one in real coordinates. At $\mu_{np} = 0$, the contribution is defined by its continuous zero limit; an arbitrary choice of phase supplies no information.

B. Geometric tangent space

For each user k and path ℓ, take the real parameters $(\psi_{\ell,k}, \Re\alpha_{\ell,k}, \Im\alpha_{\ell,k})$. Collect them in $\boldsymbol{\eta}$ and set $\mathbf{D} = \partial \mathbf{x}/\partial \boldsymbol{\eta}$. Writing $v_n(\psi) = e^{-\jmath(n-1)\psi}$, the complex derivatives for one path are

$$\frac{\partial g_k[n]}{\partial \psi_{\ell,k}} = -\jmath(n-1)\alpha_{\ell,k} v_n(\psi_{\ell,k}), \tag{54}$$

$$\frac{\partial g_k[n]}{\partial \Re\alpha_{\ell,k}} = v_n(\psi_{\ell,k}), \tag{55}$$

$$\frac{\partial g_k[n]}{\partial \Im\alpha_{\ell,k}} = \jmath v_n(\psi_{\ell,k}). \tag{56}$$

Their real and imaginary parts fill the corresponding entries of $\mathbf{x}_n$; entries belonging to other users are zero. Using spatial frequency ψ follows the channel generator directly and avoids a coordinate singularity from $\psi = \pi \sin\theta$ at endfire.

Take a thin SVD $\mathbf{D} = \mathbf{U}_D \boldsymbol{\Sigma}_D \mathbf{V}_D^T$. The implementation retains the columns of $\mathbf{U}_D$ whose singular values exceed 10^{-9} times the largest singular value. These columns form $\mathbf{U}$, an orthonormal basis for the numerically resolved range of $\mathbf{D}$. This construction avoids inverting a redundant physical parameterization when $\mathbf{D}$ is not full column rank.

For locally unbiased estimation along this tangent space,

$$\mathbf{C}_{\text{geom}} = \mathbf{U}(\mathbf{U}^T \mathbf{J} \mathbf{U})^{-1} \mathbf{U}^T. \tag{57}$$

The gains and spatial frequencies are varied jointly in $\mathbf{D}$; neither is held fixed as known side information. This calculation includes no unknown reference-field, calibration or noise-variance parameters. Adding such nuisance parameters would require the corresponding enlarged information matrix. The true path counts and parameter point specify the local reference, rather than inputs available to HS-GS.

The displayed inverses require positive definite information on the relevant space. A singular restricted matrix indicates an unresolved local direction and is not replaced by a regularized finite bound. The known reference can remove phase ambiguity, but its presence alone does not guarantee identifiability for every pilot design. Coincident or zero-gain paths can also invalidate a regular local parameterization. The stated numerical rank tolerance must therefore accompany the results.

C. Numerical evaluation and normalization

For trial i, the two scalar reference errors are

$$b_{\text{unc},i} = \sum_{n=1}^{N} \operatorname{tr}(\mathbf{J}_{n,i}^{-1}), \tag{58}$$

$$b_{\text{geom},i} = \operatorname{tr}\left[(\mathbf{U}_i^T \mathbf{J}_i \mathbf{U}_i)^{-1}\right]. \tag{59}$$

The second identity uses $\mathbf{U}_i^T \mathbf{U}_i = \mathbf{I}$. For either reference, the plotted value is

$$10 \log_{10} \frac{\sum_i b_i}{\sum_i \|\mathbf{G}_i\|_F^2}. \tag{60}$$

The same channel realizations and energy denominators are used for the estimator curves. This is a ratio of sums, not an average of trial-wise dB values.

The scalar expectation is evaluated with scaled Bessel functions and quadrature over $[\max(0, a-8\sigma), a+8\sigma]$, where $\sigma = \sqrt{\sigma^2}$. The quadrature uses absolute tolerance 10^{-14} and relative tolerance 10^{-12}, splitting at a when it lies inside the integration window. At each main-figure SNR, 800 equally spaced amplitude nodes cover $0.99a_{\min}$ to $1.01a_{\max}$ over that SNR's worlds. Piecewise cubic Hermite interpolation supplies the scalar weights at the measurement amplitudes. The implementation checks 200 sampled amplitudes against direct quadrature and requires maximum relative error below 10^{-4}. The tangent derivatives are checked by finite differences. It also checks that the constrained trace is positive and no larger than the unconstrained trace, and records the condition number of the restricted information matrix. No diagonal ridge is added.

APPENDIX B
ADDITIONAL SENSITIVITY RESULTS

A. Effect of Reference Strength

We also test how the reference strength affects the benefit of the Hankel constraint. We use $N = 32$, $K = 3$, $P = 30$, SNR $= 5$ dB, and independently sampled path counts $L_k \in \{3, \ldots, 7\}$. The single-user RSR takes the values 0, 5, 10, 12, 15, 20, and 25 dB. Both estimators use 100 final updates. HS-GS uses 25 updates per candidate rank, four Cadzow sweeps per application, and the same 30% validation split followed by a final fit using all pilots.

This supplementary analysis reuses 2,800 stored paired trials from a separate simulation campaign, with 400 trials at each RSR. These are distinct from the 27,300 main-campaign trials reported above. No new Monte Carlo realizations were generated for this analysis. We use only the classical EM-GS and HS-GS results. Pointwise 95% confidence intervals use 4,000 paired bootstrap resamples.

Fig. 6 shows that HS-GS improves on EM-GS at all seven RSR values in this experiment. The ratio-of-sums gain ranges from 1.83 to 2.51 dB, and each corresponding confidence interval is above zero. At RSR $= 0$ dB, the gain is 1.83 dB, with a 95% interval of $[1.40, 2.26]$ dB. The paired-median gain at this point is 2.83 dB; these are different summaries of the same paired errors.

The weak reference still makes channel estimation harder for both methods. At RSR $= 0$ dB, the EM-GS and HS-GS NMSEs are -6.19 and -8.01 dB. At RSR $= 12$ dB, they are -10.73 and -12.93 dB. Thus, the structural gain remains useful at the weakest tested reference, but it does not remove the loss caused by that weak reference. The gain does not increase steadily with RSR. These findings apply to the tested array size, SNR, and channel distribution.

B. Sensitivity to the Validation Fraction

We test validation fractions of 10%, 20%, 30% and 40% at $P \in \{10, 30\}$ and SNRs of 0 and 10 dB. The other settings are $N = 32$, $K = 3$, RSR $= 12$ dB and independently drawn three-to-seven paths per user. We retain the 100 final updates, 25 updates per candidate and four Cadzow sweeps. For each pilot-count/SNR pair, the four fractions use the same 400 channel, pilot and noise realizations. The validation sets are nested trailing pilot columns. All P pilots are reused for every final fit, so the comparison changes the rank-selection split rather than the total transmitted pilot count.

Fig. 7 shows that the split matters. For example, at $P = 30$ and SNR $= 10$ dB, using 10% instead of 30% gives a paired ratio-of-sums gain of -0.602 dB, with interval $[-0.832, -0.383]$ dB. At $P = 10$ and SNR $= 10$ dB, increasing the fraction from 30% to 40% gives -0.461 dB $[-0.744, -0.176]$. Negative values mean that the changed split is worse than the 30% reference. Neither a very small nor a larger validation set is therefore automatically better.

The selected ranks also change. At $P = 10$ and SNR $= 0$ dB, the median selected rank falls from three with a 10% split to one with a 40% split. Across all 16 settings, full-rank fallback occurs in 0–1.5% of the trials. HS-GS still improves on EM-GS in each setting, but the results do not identify one validation fraction that is best in all conditions.

C. Effective Rank Estimated from Measurements

The true channel is unavailable at the receiver, so the effective rank used above cannot directly control the estimator. We therefore test a simple observable proxy: calculate the effective rank from the final unstructured EM-GS estimate. For user k in trial i, define

$$\widehat{\rho}_{\mathrm{EM},k,i} = \frac{r_{\mathrm{eff}}(\mathcal{H}(\widehat{\mathbf{g}}_{\mathrm{EM},k,i}))}{r_{\max}}, \qquad \bar{\widehat{\rho}}_{\mathrm{EM},i} = \frac{1}{K}\sum_{k=1}^{K} \widehat{\rho}_{\mathrm{EM},k,i}. \tag{61}$$

We compare this trial-level mean with the corresponding mean from the true user channels and with the paired HS-GS gain. We use EM-GS estimates so that the proxy is measured before a Hankel correction changes the singular values. No true channel or path count is used to form the proxy.

This analysis reuses 10,400 existing main-campaign trials: 7,200 trials from the array-size study with $P = 30$, and 3,200 trials from the matched $N = 32$ complexity study with $P = 20$. We repeat only EM-GS using the original channels, observations, seeds, and update counts. Every reproduced EM-GS squared error agrees exactly with its stored value in this implementation. These replays are not additional independent Monte Carlo trials.

Fig. 8(a) shows that the proxy tracks true effective rank more closely as SNR increases. For $N = 32$, the Spearman correlation rises from 0.202 at -5 dB, with a 95% interval of $[0.100, 0.297]$, to 0.914 at 20 dB, with an interval of $[0.890, 0.931]$. The intervals use 4,000 bootstrap resamples of entire trials, keeping their user averages and paired errors together.

However, estimation noise raises the proxy substantially. For $N = 32$ and SNR $= 5$ dB, its mean across trials is 0.577, whereas the corresponding true-channel mean is 0.244. HS-GS still gives a positive paired-median gain of 2.544 dB. These are different summaries: the first two numbers are means of effective rank, while the last is a median of paired estimation gains. The upward shift warns against applying a cutoff near 0.5 to this proxy without calibration. This comparison of averages does not evaluate a per-trial cutoff.

We also examine the relation between the proxy and HS-GS gain across path counts. We use the matched $N = 32$, $P = 20$, RSR $= 10$ dB study and group its random SNRs into fixed 5 dB bands. In the $[-10, -5)$ dB band, the proxy–gain correlation is -0.098, with a 95% interval of $[-0.183, -0.012]$. In the $[15, 20]$ dB band, it is -0.797, with an interval of $[-0.836, -0.752]$. Negative correlation means that larger estimated effective rank tends to accompany a smaller structural gain. The true effective rank and normalized true path count are also shown as diagnostic references; neither is available to the receiver.

The proxy therefore provides some practical information, especially at higher SNR. It is not a calibrated replacement for the true effective rank, and no per-trial rank-selection rule or

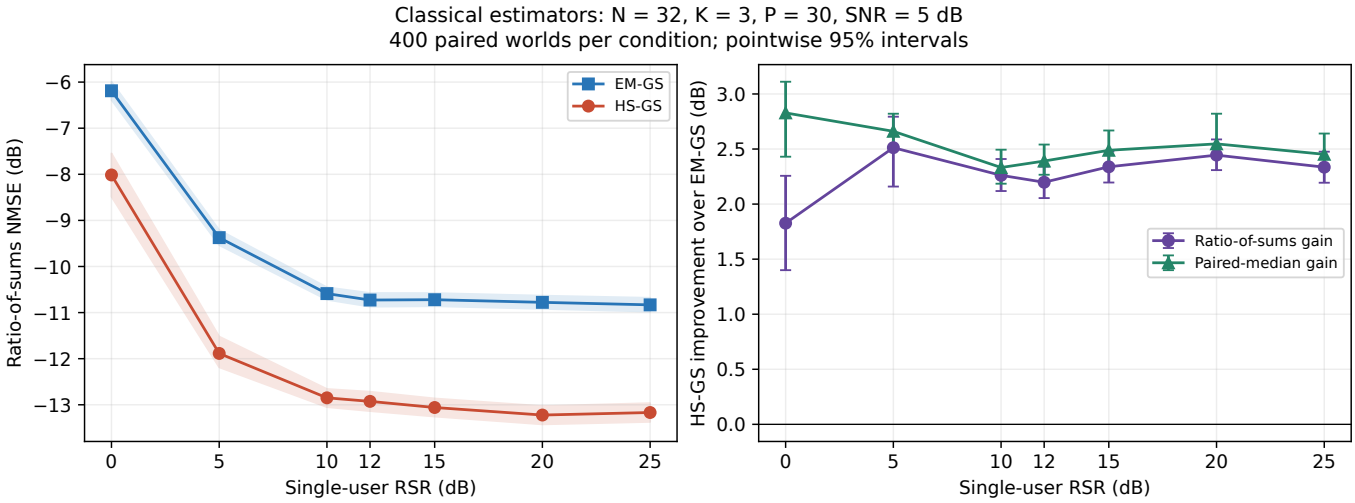


Fig. 6. Effect of single-user RSR for $N = 32$, $K = 3$, $P = 30$, and SNR $= 5$ dB. Left: ratio-of-sums NMSE. Right: HS-GS gain measured by the ratio of summed errors and by the paired median. Shaded bands and error bars are pointwise 95% paired-bootstrap intervals. Each RSR uses 400 paired trials from a separate stored campaign.

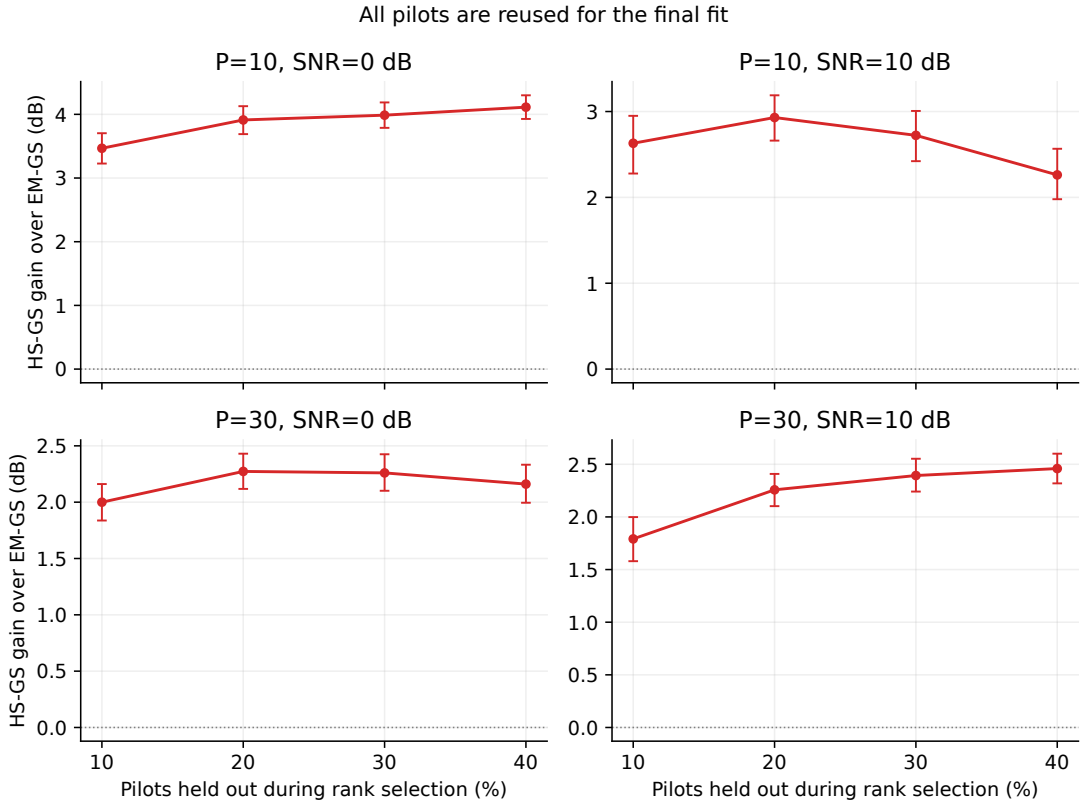


Fig. 7. HS-GS ratio-of-sums gain over EM-GS as the validation fraction changes. The fractions share 400 paired worlds within each panel, and all pilots are used for the final estimate. Error bars are pointwise 95% paired-bootstrap intervals from 4,000 resamples.

switching threshold is validated here. The SNR bands reduce, but do not remove, the influence of SNR. In this diagnostic we use means over users within each trial; the effective-rank curves in Section V-E instead summarize their operating points by medians over user columns and trials.

Appendix C
Sweep Count and Rank-Selection Stability

A. Number of structural sweeps

We use 100 fresh paired trials at each combination of $L \in \{3, 7, 14\}$ and SNR $\in \{0, 10\}$ dB, giving 600 independent channels. We keep $N = 32$, $K = 3$, $P = 30$, RSR $= 12$ dB

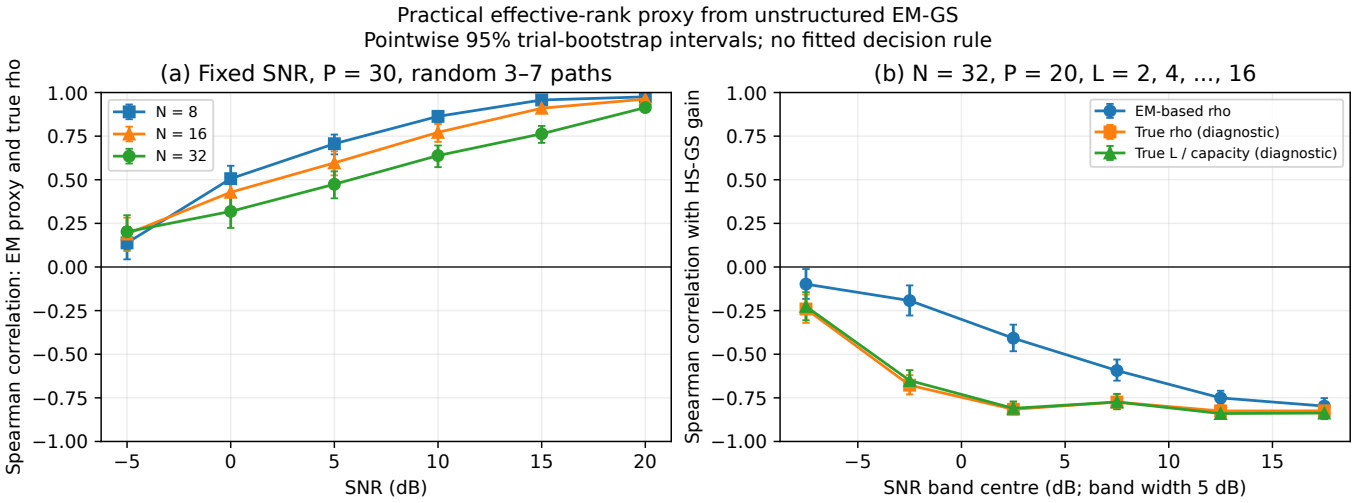


Fig. 8. Effective rank estimated from unstructured EM-GS. (a) Correlation between the observable proxy and true effective rank, using trial-level user means at each fixed SNR. (b) Correlation with paired HS-GS gain across $L \in \{2, 4, \ldots, 16\}$ within fixed 5 dB SNR bands. The two true-channel quantities are evaluation-only references. Error bars are pointwise 95% intervals from 4,000 paired trial-bootstrap resamples.

and the 100/25 update budgets. Both hard truncation and smooth filtering are tested with one, two, four and eight sweeps per application. Each setting selects its own rank or threshold using the same trailing 30% validation set, then refits all pilots. Thus, we test the whole selection-and-estimation procedure; we do not hold the selected rank fixed across sweep counts.

Fig. 9 shows that more sweeps need not improve channel estimation. One hard-rank sweep improves on four by 0.275–0.652 dB across the six settings, with all six pointwise 95% intervals above zero. For example, the gain at $L = 7$ and SNR $= 10$ dB is 0.652 dB $[0.488, 0.824]$. The original four-sweep choice is therefore not optimal in these tests. Fitting the structural constraint more closely does not necessarily estimate the noisy channel better. Because selection is repeated, this comparison includes changes in both the selected rank and the structural processing.

Smooth filtering also depends on the sweep count, but no tested count has the highest point-estimate gain in every setting. Its threshold is rescaled on each sweep, so increasing the count changes the accumulated suppression. The original comparison between hard and smooth processing must therefore be read with its stated four-sweep setting. These checks do not establish a universal best count or a stopping rule.

A separate serial check uses 12 fresh worlds at $L = 7$ and SNR $= 5$ dB, with the same array, pilots, reference and update budgets. On the same CPU in double precision with one numerical-library thread, the median hard-rank times for one, two, four and eight sweeps are 0.121, 0.201, 0.358, 0.667 s, including initialization, selection and final fitting. The corresponding smooth-filter times are 0.074, 0.118, 0.209, 0.391 s. Method order is randomly permuted within each timing world. These small timing samples describe one implementation and machine; ordinary host activity is not controlled. Parallel campaign times are not used as inference benchmarks.

TABLE IV
RANDOM VERSUS TRAILING PILOT SPLIT. EACH ROW USES 100 WORLDS; THREE RANDOM PARTITIONS ARE AVERAGED WITHIN A WORLD FOR THE GAIN.

L	SNR	Rank differs (%)	Gain [95% interval] (dB)
3	0	54.3	-0.021 $[-0.278, 0.275]$
3	10	15.7	-0.148 $[-0.333, 0.046]$
7	0	73.7	$+0.040$ $[-0.204, 0.296]$
7	10	53.7	$+0.184$ $[0.016, 0.364]$
14	0	86.7	-0.015 $[-0.225, 0.193]$
14	10	82.0	-0.071 $[-0.223, 0.087]$

B. *Changing the held-out pilots*

On the same 600 worlds, we repeat four-sweep HS-GS using three random 70/30 pilot partitions, in addition to the original trailing split. Each partition is a separate single-split run. We average the three random-partition squared errors within each world before forming the ratio-of-sums comparison and its 4,000-resample interval. The independent sample size remains 100 per setting. This is not a multi-fold estimator or an average of the resulting channel estimates.

Table IV shows no consistent performance benefit from random splitting. One pointwise interval is positive, at $L = 7$ and SNR $= 10$ dB; the other five include zero. This does not establish equivalence. Random versus fixed partitions choose different ranks in 15.7–86.7% of the paired comparisons. The small average performance differences can therefore hide substantial variation in the chosen rank. With exchangeable pilot columns, there is no preferred column position in the population model; this test mainly examines sensitivity within a particular observed world. Repeated selection adds computation and has not been shown here to reduce deployment cost.

C. *Selected rank and estimation harm*

We also reanalyse the 2,400 existing trials used for the six-method comparison. They are not additional independent

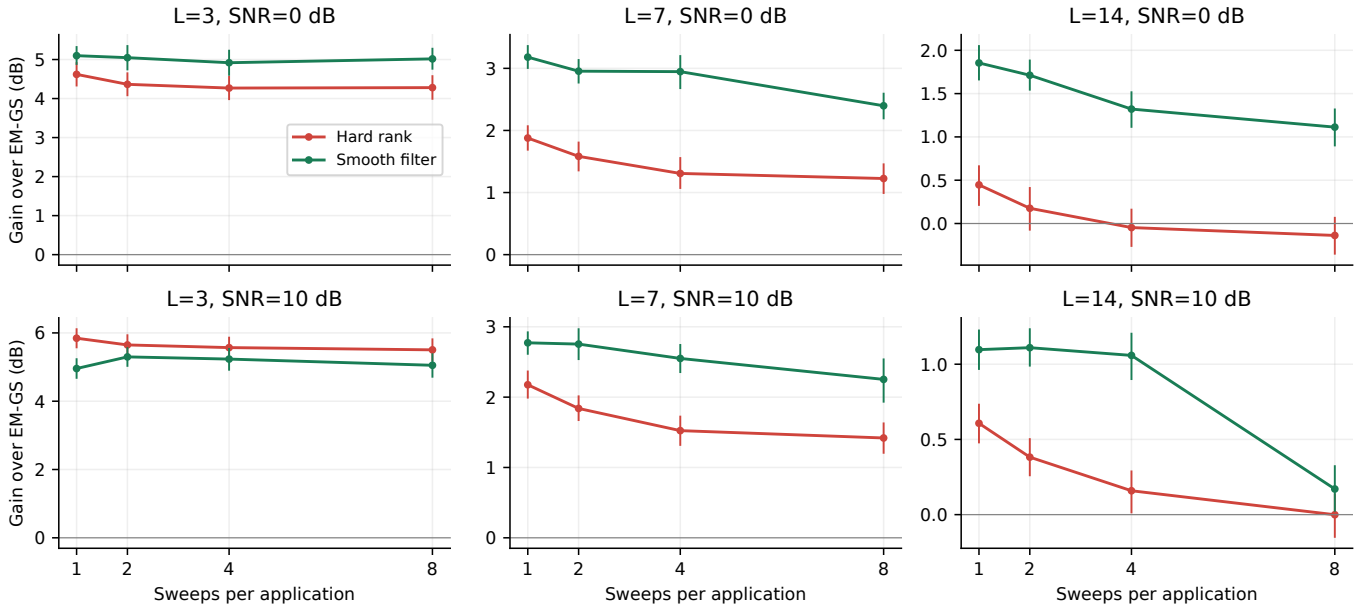


Fig. 9. Sweep sensitivity on 100 fresh paired trials per panel. Each method and sweep count selects its own rank or threshold. Gain is the difference between ratio-of-sums NMSE values; positive values favor the structured method over EM-GS. Bars are pointwise 95% intervals from 4,000 paired world resamples.

TABLE V
RANK AND HARM DIAGNOSTICS FROM THE EXISTING 400 TRIALS PER ROW. HARM MEANS STRICTLY LARGER SQUARED ERROR THAN EM-GS.

L	SNR	Median $\widehat{r}$	$\widehat{r} < L$ (%)	Harm (%)
3	0	3	39.00	0.25
3	10	3	1.75	0.25
7	0	5	78.75	8.75
7	10	7	49.25	4.00
14	0	7	92.50	47.75
14	10	11	79.00	27.50

trials. These cells have the same path count for every user; under the stated distinct-path model, the noiseless Hankel rank is L. Table V reports how often the shared selected rank is below L and how often HS-GS has larger squared channel error than its paired EM-GS estimate.

Selecting below the true rank is not automatically a failure. At $L = 7$ and SNR $= 0$ dB, 315 of 400 trials select below seven, but only 31 of those 315 trials are worse than EM-GS. Suppressing weak components can help in noise. Conversely, at $L = 14$ and SNR $= 0$ dB, HS-GS is worse in 191 of 400 trials. The full-rank option does not prevent the selector from choosing a harmful smaller rank. These are descriptive associations, not proof that matching the true rank minimizes channel error.

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